
EE5606/AI5120 Convex Optimization, Spring 2026
Indian Institute of Technology, Hyderabad
Homework 1; Convex Optimization
Total Marks: **50 points**
Assigned on 09/04/2026, Due date **11:59PM on 17/06/2026**

Instructions

- It is strongly recommended that you work on your homework on an individual basis. If you have any questions or concerns, feel free to talk to the instructor or the TAs.
 - Use `matplotlib` to read and plot images.
 - Use `numpy` for basic functions like `log`, `sqrt`, `power`. Do not use other built-in functions.
 - Please turn in Python Notebooks with the following notation for the file name: your-roll-number-hw1.ipynb.
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Question 1

(10 points)

- (a) Define a **quasiconvex function**.
- (b) Check whether the function

$$f(x) = \begin{cases} 1, & x < 0 \\ 0, & x = 0 \\ 1, & x > 0 \end{cases}$$

is quasiconvex.

- (c) Is this function convex? Justify your answer.
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Question 2

(10 points)

- (a) Define the **conjugate function** $f^*(y)$.
- (b) Find the conjugate of

$$f(x) = e^{-x}, \quad x \in \mathbb{R}.$$

Question 3

(10 points)

- (a) Define the **sublevel set** S_α .



(b) For

$$f(x_1, x_2) = \sqrt{x_1^2 + x_2^2} - x_1, \quad (x_1, x_2) \in \mathbb{R}^2,$$

determine whether f is quasiconvex by analyzing S_α .

Question 4

(10 points)

(a) Define a **log-concave function** and a **log-convex function**.

(b) Check whether

$$f(x) = \frac{1}{1 + x^2}, \quad x \in \mathbb{R}$$

is log-concave or log-convex (or neither).

Question 5

(10 points)

(a) State the **first-order condition for quasiconvexity** (for differentiable f).

(b) Let

$$f(x) = x^4, \quad x \in \mathbb{R}.$$

- Show that f is convex.
- Find its conjugate $f^*(y)$.
- Interpret the condition $y = \nabla f(x)$ in terms of the conjugate.